4.13 Solution: By symmetry, the electric field between the conducting surfaces should have the following form,

$$\mathbf{E} = \frac{A}{\rho}\hat{\rho},$$

where the constant A can be determined by

$$\int_a^b E_\rho d\rho = V,$$

which leads to

$$A = \frac{V}{\log(b/a)}.$$

The gravitational potential energy required in raising the dielectric liquid to an average height of h is

$$\pi(b^2 - a^2) \cdot \rho g h^2,$$

which is provided by the electric energy,

$$\frac{1}{2}\int \mathbf{P} \cdot \mathbf{E} d^3x = \frac{\varepsilon_0 \chi_e}{2} \cdot 2\pi h \int_a^b \frac{A^2}{\rho^2} \cdot \rho d\rho = \varepsilon_0 \chi_e \pi h \cdot \frac{V^2}{\log(b/a)}.$$

Therefore,

$$\chi_e = \frac{(b^2 - a^2)\rho g h \log(b/a)}{\varepsilon_0 V^2}.$$